

Lab 6 ticket

Impossible Equations — Friday, December 4

Name: _____ Language (circle): Octave / Python

New syntax (fill as you go)

I needed to...	Command / pattern (your language)
Least-squares solve $\hat{x}$	
Fitted $\hat{b} = A\hat{x}$	
Residual / RMS	
Check $A^T r$	

Quiz 10 today is Week 13 eigenvalues (**not** this lab). SVD / capstone launch share the period; Lab 6 is least squares only.

E1. $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, $b = (1, 2, 2)$. No exact sol.? $\hat{x}$, $\|r\|$.

Expected: $\hat{x} \approx (1.167, 0.500)$; $\|r\| \approx 0.408$. **Yours:** $\hat{x} =$ _____ $\|r\| =$ _____

E2. Focus scores $b = (3, 4.5, 4, 6.5, 7)$ vs $t = 1..5$; model $x_1 + x_2 t$. Record $\hat{x}$ and RMS.

Expected: $\hat{x} = (2, 1)$; RMS ≈ 0.55 . **Yours:** $\hat{x} =$ _____ RMS = _____

Useful on a 0–10 scale? (one phrase) _____

E3. Plot data + fitted line. Trend despite $t = 3$ dip? _____

E4. $A^T r \approx$ _____ Why near 0? _____

E5. Two checks beyond quoting $\hat{x}$: (1) _____ (2) _____

If RMS = 3.5 on 0–10 scale: useful? _____

Stuck on (optional): E _____

Instructor initials: _____